

Sri Sai Charan Yarlagadda

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EDUCATION

Northeastern University

Boston, MA | Sept 2024 – Dec 2026

Masters in Data Science, GPA: 3.73

Coursework: Supervised Machine Learning, Data Mining and Modeling, Computer Vision, MLOps, Data Engineering and Management

Kalinga Institute of Industrial Technology

Bhubaneswar, India | Sept 2020 – May 2024

B.Tech. in Computer Science Engineering, GPA: 3.76

Coursework: Machine Learning, Deep Learning, Data Analytics, Data Structures and Algorithms, Cloud Computing

EXPERIENCE

ML Engineer / Data Scientist

Brooklyn, NY | Aug 2025 – Dec 2025

Crewasis.ai

- Extracted 10K+ publication records from Google Scholar via SerpAPI and ingested them into S3 alongside 15+ domain-specific corpora through Lambda-triggered pipelines, then designed LangChain workflows using ConversationalRetrievalChain and OutputFixingParser for structured JSON extraction, achieving 91.7% accuracy across 2,000+ monthly documents and reducing manual review by 40%
- Fine-tuned Claude Sonnet 3.7 using QLoRA adapters via AWS Bedrock on the ingested domain corpus for enterprise Q&A, and deployed Llama 3.3 70B (4-bit GPTQ quantized) on EC2 G5 instances with vLLM serving for batch document summarization, improving domain accuracy by 20% and reducing inference latency by 30%
- Trained RNN-based sequence models (GRU, LSTM) and Transformer encoder layers with self-attention mechanisms using PyTorch on operational data for sequential pattern recognition, feeding model outputs into the personalized LLM module to generate knowledge-base-driven AI insights with 20% improved relevance
- Developed classification (Random Forest, XGBoost) and regression (Lasso, Ridge) models using Scikit-learn on tabular operational data with 2,000+ monthly records, performing stratified splits and recursive feature elimination to optimize prediction accuracy for downstream tasks
- Orchestrated the full pipeline on AWS with S3 for corpus storage, Lambda for event-driven ingestion, EC2 for GPU model serving, Bedrock for managed LLM inference, and CloudWatch for monitoring, improving reliability by 35% and reducing end-to-end latency by 45%

Data Science / ML Intern

Ahmedabad, India | Jul 2023 – Nov 2023

BrainerHub Solutions LLP

- Built 3 client-facing application modules: a stock forecasting engine using ARIMA for trend baselines and LSTM for nonlinear sequence capture, a lead scoring classifier (XGBoost, 82% precision), and a recommendation engine using collaborative filtering on 50K+ interaction records, achieving 15% average KPI uplift
- Developed a Bi-LSTM intent classification model using PyTorch on 10K+ labeled text samples with GloVe embeddings for enterprise chatbot routing, integrating it as an upstream router feeding into GPT-3.5 Turbo via the OpenAI API for automated client report generation, reducing manual drafting time by 25%
- Performed end-to-end feature engineering on client datasets using Pandas, NumPy, and Scikit-learn, creating 20+ features including TF-IDF text embeddings, one-hot categoricals, and binned numerical ranges, resolving 8% missing data through KNN imputation
- Deployed application modules on AWS EC2 with Docker and GCP Cloud Run for auto-scaling, tracking experiments with MLflow and creating Power BI dashboards with 10+ visualizations for weekly client reporting

PROJECTS

Pneumonia Detection – CNN and Transfer Learning

Jan 2026 – Apr 2026

- Fine-tuned a ResNet50 CNN pre-trained on ImageNet using PyTorch on the ChestXRay2017 dataset (5,216 training images), replacing the final FC layer for binary classification and addressing 3:1 class imbalance with weighted random sampling
- Built a modular C++ inference pipeline using OpenCV for image loading, BGR-to-RGB conversion, bilinear resize to 224x224, ImageNet channel-wise normalization, and HWC-to-CHW tensor layout, integrated with ONNX Runtime for forward pass achieving 99.49% recall, 88.89% F1-score, and 0.199s per-image inference on 624 test images
- Analyzed precision-recall trade-offs for clinical screening (388 TP, 2 FN, 95 FP, 139 TN) and built an OpenCV-based visualization module annotating X-rays with confidence scores and color-coded prediction banners for qualitative error analysis

Credit Card Fraud Detection – Anomaly Detection

Mar 2025 – Apr 2025

- Applied PCA and JSD-based feature selection on 284,807 transaction records, reducing dimensionality by 42% while preserving anomaly signal through statistical correlation analysis, improving model training efficiency by 38%
- Trained Isolation Forest for unsupervised anomaly scoring and autoencoder reconstruction-error models alongside a DNN classifier with SMOTE resampling using TensorFlow/Keras, achieving ROC-AUC 0.91 and 93.2% precision on 0.17% fraud-rate data
- Conducted 5-fold stratified cross-validation with grid search across 50+ configurations, tracking precision, recall, and F1-score with less than 2% variance across folds to validate robustness on unseen transaction patterns

Customer Churn – Predictive Classification on Azure

Apr 2025 – Jun 2025

- Extracted 500K+ records from Snowflake via optimized SQL (CTEs, window functions) and orchestrated ETL through Azure Data Factory, engineering 8+ RFM and cohort features with 92% data quality through automated validation
- Trained K-means clustering (8 segments), XGBoost, and a 3-layer DNN on Azure ML Studio with experiment tracking via MLflow, achieving AUC 0.86 with 83% precision and enabling 21% retention uplift
- Designed Tableau dashboards with 15+ interactive visualizations of churn drivers, segment profiles, and feature importance for non-technical stakeholders

Walmart Sales Forecasting – Time Series and Foundation Models

Dec 2023 – Mar 2024

- Cleaned 420K+ retail time-series records across 45 stores, performing Augmented Dickey-Fuller stationarity tests and STL seasonal decomposition to validate trend, seasonality, and residual components before model fitting
- Benchmarked ARIMA and SARIMAX for linear trend modeling, Bi-LSTM for capturing long-range sequential dependencies, and pre-trained foundation models (Chronos, TimesFM) for zero-shot forecasting using expanding-window cross-validation, achieving 11% lower RMSE and comparing accuracy, data requirements, and inference cost across families
- Applied Monte Carlo simulation for demand uncertainty quantification and produced 8+ visualizations of forecast distributions, seasonal patterns, and prediction intervals using Matplotlib and Seaborn for operational inventory planning

TECHNICAL SKILLS

Languages: Python, C++, SQL, R, JavaScript

ML Frameworks: PyTorch, TensorFlow, Scikit-learn, Keras, XGBoost, LightGBM, Statsmodels, SciPy

Methods: Supervised and Unsupervised Learning, Deep Learning (CNN, RNN, LSTM, Transformer), Time Series Forecasting, Foundation Models (Chronos, TimesFM), Anomaly Detection (Isolation Forest, Autoencoders), Transfer Learning, Sequence Modeling, Attention Mechanisms

AI and NLP: LangChain, HuggingFace Transformers, Prompt Engineering, RAG Architecture, Fine-tuning (LoRA, QLoRA)

Data Processing: Pandas, NumPy, PySpark, SQL (PostgreSQL, MySQL, Snowflake), OpenCV, ONNX Runtime

Cloud and MLOps: AWS (S3, Lambda, EC2, SageMaker, Bedrock), Azure (ML Studio, Data Factory, Blob Storage), GCP, Docker, Git, CI/CD, MLflow, Weights and Biases, Airflow

Visualization: Tableau, Power BI, Matplotlib, Seaborn, Plotly, Streamlit